Massyle DENDENE

PhD Candidate in Applied Mathematics
École polytechnique · Institut Polytechnique de Paris
massyle.dendene@polytechnique.edu · +33 7 67 67 67 67
massyle-d.github.io · LinkedIn: massyle-dendene · Palaiseau, France

Research Profile

My research focuses on stochastic control of self-exciting point processes, particularly Hawkes processes, with applications to cyber risk modelling in insurance and finance. Alongside my research, I teach probability and financial mathematics at ENSAE Paris.

Education

PhD in Applied Mathematics 2025–2028 (expected)
École polytechnique, Institut Polytechnique de Paris – ENSAE–CREST
Thesis topic: Intensity control for Hawkes processes.
Supervisors: Roxana Dumitrescu and Caroline Hillairet. *Co-advisor:* Nicolas Baradel.

MSc in Probability and Finance 2024–2025
Université Paris Cité, Paris
Stochastic control and calculus, backward stochastic differential equations, partial differential equations, numerical methods and Monte Carlo simulation.

Engineering Degree – Major in Actuarial Science 2023–2025
ENSAE Paris, Institut Polytechnique de Paris, Palaiseau

Bachelor's Degree in Mathematics and Economics 2022–2023
Université de Bretagne Occidentale, Brest

Engineering Studies – Major in Applied Mathematics 2021–2023
IMT Atlantique, Brest

Preparatory Classes for Grandes Écoles – PCSI-PSI* 2019–2021
Lycée Jean-Baptiste Say, Paris

Publications

Inconming

Talks and Conferences

SIAM Conference on Financial Mathematics and Engineering – Upcoming 15–18 June 2027
Washington, DC, USA

FiME PhD Seminar – Upcoming 18 December 2026
Institut Henri Poincaré, Paris, France

Séminaire Actuariat – Upcoming 20 November 2026
ISFA, Lyon, France

Research and Professional Experience

Quantitative Research Intern May–October 2025
Exiom Partners, Paris

- Developed a fractional jump-diffusion model for European option pricing in illiquid markets, using subordinated Lévy processes.
- Solved a fractional Dupire PIDE involving Caputo derivatives and validated the numerical scheme against Monte Carlo simulations.

- Analysed the impact of illiquidity on implied volatility surfaces and integrated the model into an economic scenario generator for life insurance.

Actuarial dissertation: Illiquidity risk: from European option pricing to its integration into economic scenarios for life insurance (in French).

Admitted as a member of the Institut des actuaires following the successful defence of this dissertation.

Non-Life Actuarial Intern June–September 2024

Aon Global Risk Consulting, Luxembourg

- Reserving for non-life reinsurance captives using Chain-Ladder, Bornhuetter–Ferguson and bootstrap methods.

Research Intern – Supervisor: Stéphane Loisel April–August 2023

SAF Laboratory, ISFA, Université Claude Bernard Lyon 1

- Developed statistical methods for longevity model selection and structural change detection in mortality data (AXA Joint Research Initiative).
- Modelled and mapped hail trajectories in Python, incorporating hailstone size data.

Teaching Experience

Teaching Assistant – ENSAE Paris 2025–Present

Institut Polytechnique de Paris, Palaiseau

115 hours of tutorials, from third-year undergraduate to second-year master's level (L3–M2).

- Probability Theory.
- Financial Mathematics and Financial Instruments.
- Introduction to Stochastic Processes.

Led tutorials, supported students and graded assessments.

Private Mathematics Tutor

Secondary school and classes préparatoires

Individual support in course material and problem solving; preparation for written and oral competitive examinations, with an emphasis on rigorous reasoning and independent work.

Skills and Languages

Mathematics	Probability, stochastic calculus and control, statistics, optimisation, numerical methods and time series.
Programming	Python, R; Git, LaTeX.
Languages	French (native), English (C1; IELTS 7/9), German (B2), Italian (A2).

Interests and Activities

Piano: ten years of conservatory training; concerts including one at the Philharmonie de Paris.

Table tennis: competitive play; organisation of weekly training sessions for the student club.

Other: climbing; regular blood donation.